# AAKRITI JAIN

aakriti.jain@utdallas.edu | +1 (332) 248 7111 | aakriti-jain.github.io
Google Scholar: scholar.google.com/citations?user=qqWOQgIAAAAJ

## EDUCATION

**Ph.D., Computer Science** *(GPA: 4.00/4.00)* Expected May 2027
The University of Texas at Dallas, USA
Advisor: Prof. Abusayeed Saifullah

**M.S., Computer Science** *(GPA: 4.00/4.00)* 2024
Wayne State University, USA

**B.Tech., Electronics and Communications Engineering** *(CGPA: 9.3/10)* 2021
ADGITM, Guru Gobind Singh Indraprastha University, New Delhi, India

## RESEARCH INTERESTS

Internet of Things (IoT) • Real-Time and Embedded Systems • Wireless Networks • Low-Power Wide-Area Networks (LPWANs) • Cyber-Physical Systems (CPS) • Energy-Aware Systems

**Research Summary.** I build low-power wireless systems that enable timely, reliable, and energy-efficient communication for wide-area industrial IoT and CPS, from oil fields and smart grids to remote process plants. In LPWANs, thousands of energy-constrained devices contend for the same channel under random access, which makes meeting stringent deadlines within tight energy budgets a fundamental challenge. My research brings real-time operation to LPWANs such as LoRa and LR-FHSS through collision- and deadline-aware medium access control (MAC) protocols that remain reliable even under bursty traffic, energy-aware communication for intermittently powered devices, and scheduling–control co-design that treats the network as part of the control loop. My broader work encompasses latency-aware data collection using mobile sinks, cross-technology communication from LR-FHSS to LoRa, and ML-based scheduling for multicore processors. I validate these designs on software-defined radio (SDR) prototypes with GNU Radio and USRP B200, multi-node COTS LoRa testbeds, and large-scale simulation and optimization in NS-3, PyTorch, and Gurobi. Next, I aim to bring these guarantees to direct-to-satellite and 5G/6G IoT.

## PUBLICATIONS

*Venues: RTSS, SenSys, RTAS, ICCPS, DCOSS-IoT, IoTDI, TCPS, and TECS.*
*Each title links to its full text.* $^*$*Equal contribution*

### CONFERENCE PAPERS

[C7] Md Ashikul Haque, **Aakriti Jain**, Venkata Prashant Modekurthy, Abusayeed Saifullah. *Enabling Cross Technology Communication from LR-FHSS to LoRa.* In Proc. ACM/IEEE International Conference on Embedded Artificial Intelligence and Sensing Systems (**SenSys 2026**), CPS-IoT Week 2026, pp. 102-114. Acceptance rate: 19.8%.

[C6] Abusayeed Saifullah, Nasif Ahmed, **Aakriti Jain**, Md Yusuf Sarwar Uddin, Mohammad Ashiqur Rahman. *Latency-Optimized Data Harvesting in LoRa Networks Using a Mobile Sink.* In Proc. IEEE Real-Time and Embedded Technology and Applications Symposium (**RTAS 2026**), pp. 360-372. Acceptance rate: 25.2%.

[C5] **Aakriti Jain**, Venkata Prashant Modekurthy, Abusayeed Saifullah, Marco Brocanelli. *Co-Scheduling of Charging and Transmission in Supercapacitor Powered IoT.* In Proc. IEEE International Conference on Distributed Computing in Smart Systems and the Internet of Things (**DCOSS-IoT 2026**), pp. 137-144. Acceptance rate: 23.9%.

[C4] **Aakriti Jain**, Md Ashikul Haque, Abusayeed Saifullah, Haibo Zhang. *Burst-MAC: A MAC Protocol for Handling Burst Traffic in LoRa Network.* In Proc. IEEE Real-Time Systems Symposium (**RTSS 2024**), pp. 148-160. Acceptance rate: 22.8%.

[C3] **Aakriti Jain**$^*$, Venkata Prashant Modekurthy$^*$, Abusayeed Saifullah. *Control over Low-Power Wide-Area Networks.* In Proc. ACM/IEEE International Conference on Cyber-Physical Systems (**ICCPS 2024**), CPS-IoT Week 2024, pp. 192-201. Acceptance rate: 23%.

[C2] Mohammad Pivezhandi, **Aakriti Jain**, Abusayeed Saifullah, Ali Jannesari. *Work-In-Progress: Energy and Thermal-Aware Scheduling based on HMARL for OpenMP DAG Workloads.* In Proc. IEEE Real-Time Systems Symposium (**RTSS 2024**), Brief Presentations Track, pp. 455-458.

[C1] Sezana Fahmida, Venkata Prashant Modekurthy, Dali Ismail, **Aakriti Jain**, Abusayeed Saifullah. *Real-Time Communication over LoRa Networks.* In Proc. ACM/IEEE International Conference on Internet-of-Things Design and Implementation (**IoTDI 2022**), CPS-IoT Week 2022, pp. 14-27.

## JOURNAL ARTICLES

[J2] **Aakriti Jain** *, Venkata Prashant Modekurthy *, Abusayeed Saifullah. *Real-Time Scheduling and Control Co-Design over LPWAN.* ACM Transactions on Cyber-Physical Systems (**TCPS**), 2026, 26 pages.

[J1] Sezana Fahmida*, **Aakriti Jain***, Venkata Prashant Modekurthy*, Dali Ismail, Abusayeed Saifullah. *RTPL: A Real-Time Communication Protocol for LoRa Network.* ACM Transactions on Embedded Computing Systems (**TECS**), 2024, 31 pages.

## MANUSCRIPTS UNDER REVIEW

[U2] **Aakriti Jain**, Abusayeed Saifullah, Haibo Zhang. *Random or Deterministic? Coordinating Frequency and Time for Collision Avoidance in LR-FHSS.* Under review at **ACM MobiCom 2027**.

[U1] **Aakriti Jain**, Abusayeed Saifullah, Haibo Zhang. *Beyond Random Access: Structured Subcarrier Grouping for LR-FHSS.* Under review at **IEEE PerCom 2027**.

## SELECTED TALKS AND POSTERS

*Traffic-Aware Protocols for Smart Wireless Networks* — Jan 2025
Invited Talk, Wayne State University, Detroit, MI, USA

*Burst-MAC: A MAC Protocol for Handling Burst Traffic in LoRa Network* — Dec 2024
Paper Talk, IEEE RTSS 2024, York, UK

*Energy- and Thermal-Aware Scheduling Based on HMARL for OpenMP DAG Workloads* — Dec 2024
Poster, IEEE RTSS 2024, York, UK

*Control over Low-Power Wide-Area Networks* — May 2024
Paper Talk, ACM/IEEE ICCPS 2024, Hong Kong, China

## RESEARCH EXPERIENCE

**Graduate Research Assistant, Dept. of Computer Science** — 2021 – 2026
Wayne State University, USA (Fall 2021 – Summer 2025) and The University of Texas at Dallas, USA (Fall 2025 and Summer 2026) | Advisor: Prof. Abusayeed Saifullah

- **Coordinated Access for LR-FHSS Networks (ongoing; first-author).** In LR-FHSS, devices split packets into fragments and hop across subcarriers using random hopping patterns, so fragment collisions grow as networks become denser. I am developing structured subcarrier grouping and coordinated frequency–time scheduling as alternatives to purely random access [U1, U2; under review]. Developed a Python-based LR-FHSS network simulator and a GNU Radio gateway using USRP B200 for evaluation.
- **Real-Time and Collision-Aware MAC Protocols for LPWANs.** Designed Burst-MAC to handle bursty LoRa traffic, achieving up to 4.5× higher packet reception and 6× lower energy consumption [C4]. Developed RTPL, a real-time communication protocol for LoRa, improving deadline satisfaction by 75% without sacrificing throughput or energy [J1, C1]. Evaluated through large-scale NS-3 simulation and physical LoRa testbeds.
- **Scheduling–Control Co-Design for Industrial CPS over LPWANs.** Developed scheduling and control co-design framework with adaptive sensor sampling to optimize control performance while ensuring stability and real-time constraints. Established, to our knowledge, the *first* sufficient schedulability condition for LoRa networks. The co-design framework reduced control cost by at least 80% [J2, C3].
- **Charging–Transmission Co-Scheduling for Battery-Free IoT.** Developed a multi-objective ILP for joint charging and transmission scheduling in supercapacitor-powered IoT, accounting for stochastic energy harvesting, self-discharge, and time-sensitive packet utility. Developed LP-relaxation-based approximation and lightweight online heuristic, improving throughput and packet utility by 10–25% over baselines [C5].
- **Mobile-sink data harvesting.** Contributed to a drone-assisted data collection framework for remote LoRa deployments, establishing a 2-approximation latency guarantee and reducing collection latency by up to 80% and energy consumption by 30% [C6].
- **Cross-technology and satellite communication.** Co-developed, to our knowledge, the *first* cross-technology communication (CTC) from LR-FHSS to LoRa, per-message delivery certificates, and signature-bound frame recovery for direct-to-satellite LR-FHSS on GNU Radio and USRP [C7].
- **Learning-based multicore scheduling.** Contributed hierarchical multi-agent reinforcement learning for energy- and thermal-aware scheduling of OpenMP workloads on multicore platforms [C2].

## AWARDS AND HONORS

**Michael E. Conrad Graduate Research Award (Runner-Up)** 2025
Wayne State University

**Dean's List Award, top 2% of graduating class** 2024
College of Engineering, Wayne State University

**Graduate Student Professional Travel Award (two-time recipient)** 2023, 2024
Wayne State University

## TEACHING EXPERIENCE

**Graduate Teaching Assistant, Dept. of Computer Science** Spring and Fall 2026
The University of Texas at Dallas, USA
Course: CS 5390 Computer Networks

- Graded assignments, exams, and project deliverables, providing constructive feedback to support student learning.
- Mentored students through course projects, clarifying technical requirements and guiding implementation, troubleshooting, and design decisions.
- Helped students understand course concepts through office hours, technical discussions, and individualized student support.

**Guest Lecturer, Dept. of Computer Science** 2025
The University of Texas at Dallas, USA
Course: Internet of Things (graduate)

- Delivered multiple lectures on LoRa networking and network simulation in NS-3.

**Undergraduate Teaching Assistant, Dept. of Electronics and Communication** 2018 – 2020
ADGITM, Guru Gobind Singh Indraprastha University, New Delhi, India
Courses: ETEC 405 Wireless Networks, ETEC 310 Data Communication and Networks, ETEC 305 Microprocessors and Microcontrollers, ETCS 260 Computer Organization and Architecture.

- Assisted faculty across four semesters in delivering course materials, preparing lecture notes, and supporting laboratory instruction and student learning.

## MENTORING

**Nasif Ahmed, Ph.D. student, The University of Texas at Dallas** Fall 2025 – Present
Junior doctoral student in our laboratory. Mentored him on identifying new problems and approaching their solutions, on experiments with COTS LoRa devices, and on writing systems papers; co-authored [C6].

**Punith Jayaramu, M.S. student, Wayne State University** Fall 2024
Guided his transition from foundational wireless communication concepts to independent research, co-developed a Python-based LR-FHSS network simulator, and mentored him on writing his M.S. thesis.

**Arefa Haque, M.S. student, Wayne State University** Summer 2024
Guided her in implementing LR-FHSS on COTS devices and co-developed a GNU Radio gateway from scratch using a USRP B200, training her in experimental design, hardware debugging, and end-to-end system validation.

## PROFESSIONAL SERVICE

**Journal Reviewer**

- ACM Transactions on Cyber-Physical Systems (TCPS) 2025 – 2026
- ACM Transactions on Embedded Computing Systems (TECS) 2026
- ACM Transactions on Sensor Networks (TOSN) 2024
- Pervasive and Mobile Computing (Elsevier) 2023 – 2025

**Conference Sub-Reviewer**

- ACM/IEEE Intl. Conf. on Embedded Artificial Intelligence and Sensing Systems (SenSys) 2024 – 2027
- IEEE Real-Time Systems Symposium (RTSS) 2023 – 2026
- IEEE Real-Time and Embedded Technology and Applications Symposium (RTAS) 2022 – 2023
- Intl. Conf. on Embedded Wireless Systems and Networks (EWSN) 2023 – 2025

- IEEE Intl. Conf. on Distributed Computing in Smart Systems and the IoT (DCOSS-IoT) 2023 – 2026
- IEEE Intl. Conf. on Sensing, Communication, and Networking (SECON) 2026
- IEEE Intl. Conf. on Embedded and Real-Time Computing Systems and Applications (RTCSA) 2024 – 2025
- International Conference on Networking, Systems and Security (NSysS) 2025

## RESEARCH SUPPORT

My doctoral research has been supported by seven National Science Foundation (**NSF**) awards (CAREER-2306486, CNS-2301757, CNS-2306745, CNS-2211640, CNS-2601685, CNS-2602744, and CNS-1948365) and one Office of Naval Research (**ONR**) award (N00014-23-1-2151), with **Prof. Abusayeed Saifullah** as principal investigator and collaborations spanning more than eight universities.

## PRIOR EXPERIENCE

**Intelligent Transportation Systems Intern, Delhi Metro Rail Corporation** Jun 2019 – Aug 2019
New Delhi, India

- Developed a CNN-based passenger-classification module for Delhi Metro station entry surveillance pipeline. Analyzed smart-card transactions using clustering and sequence mining to identify passenger travel patterns for transit planning. Gained hands-on experience with RFID, NFC, and EMI subsystems for automated fare collection.

**Undergraduate Research Assistant, ADGITM, GGSIP University** Sep 2018 – Feb 2019
Delhi, India

- Led a four-person team building a smart IoT water-level monitoring and control system with a custom multilevel resistive sensor, touch-switch inlet control, and a mobile app for remote monitoring.

**Instrumentation and Data Analytics Intern, Indraprastha Gas Limited** Jun 2018 – Aug 2018
New Delhi, India

- Developed Random Forest and XGBoost models to detect flow-meter faults caused by sensor aging, environmental drift, and calibration errors, improving detection accuracy by 17% over threshold-based checks. Delivered a predictive-maintenance interface to help field teams identify instrumentation faults.

## TECHNICAL SKILLS

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|---|---|
| **Programming:** | Python, C, C++, MATLAB, SQL, R, Java, Bash |
| **Wireless and SDR:** | GNU Radio, USRP B200, COTS LoRa/LR-FHSS nodes and gateways |
| **Learning and Simulation:** | PyTorch, Pandas, NS-3 |
| **Systems and Tools:** | Gurobi, Linux, Docker, Git, LaTeX |

## REFERENCES

**Dr. Abusayeed Saifullah (advisor)**
Associate Professor, Dept. of Computer Science
The University of Texas at Dallas
abusayeed.saifullah@utdallas.edu
+1 (972) 883-4171

**Dr. Venkata Prashant Modekurthy**
Assistant Professor, Dept. of Computer Science
University of Nevada, Las Vegas (UNLV)
prashant.modekurthy@unlv.edu
+1 (702) 774-3405

**Dr. Haibo Zhang**
Associate Professor
School of Computer Science and Engineering
UNSW Sydney, Australia
haibo.zhang@unsw.edu.au

**Dr. Marco Brocanelli**
Assistant Professor, Computer Science and Engineering
The Ohio State University
brocanelli.1@osu.edu